



وزارة الاتصالات
وتكنولوجيا المعلومات



9-Month Diploma

Solar Power Generation Forecasting

Machine Learning and Deep Learning
for 15-Minute and 24-Hour-Ahead Photovoltaic Forecasting

Track: Applied AI & Data Analytics.

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Introduction: Renewable Energy Context

- Renewable energy comes from naturally replenished sources such as solar, wind, and hydropower.
- Solar power is expanding because it is clean, scalable, and suitable for residential, commercial, industrial, and utility-scale generation.
- However, solar generation is variable and weather-dependent, creating a strong need for accurate forecasting.

**Forecasting makes renewable power
more predictable and easier to integrate into the grid.**

Introduction: Photovoltaic Generation

- Solar photovoltaic systems convert sunlight directly into electrical power.
- PV output depends mainly on:
 - 1.Solar irradiance
 - 2.Temperature and local weather conditions
 - 3.System characteristics and plant configuration
 - 4.Location and solar geometry

Solar energy is abundant, but its output changes continuously across time and operating conditions.

Introduction: Forecasting Value

Forecasting Makes Solar Power Grid-Ready:

- Solar power forecasting estimates future PV generation using:
 1. Historical power measurements
 2. Weather observations or forecasts
 3. Solar-position and plant information
- Accurate forecasts help operators balance supply and demand, schedule energy resources, reduce curtailment, and integrate renewable generation more reliably.

**Better forecasts lead to
better energy planning and more reliable renewable integration.**

Problem Definition: Weather Uncertainty

Weather Variability Drives Forecast Uncertainty:

- Cloud Cover causes sudden irradiance drops and rapid fluctuations in PV output.
- Temperature and Humidity affect PV conversion efficiency and atmospheric light transmission.
- Solar Irradiance changes with time, season, location, and atmospheric conditions.

Core problem: Historical power alone cannot fully explain future output when weather conditions change quickly.

Problem Definition: Temporal Complexity

Solar Generation Has Multiple Time Scales:

Daily Cycles

Generation rises during daylight and falls to zero at night.

Seasonal Patterns

Irradiance and production vary across seasons.

Rapid Changes

Cloud shadows can create sudden intra-hour power drops.

Long-Term Effects

Panel degradation and changing operating conditions affect generation over time.

The model must capture patterns across multiple time scales.

Problem Definition: Operational Risk

Forecast Errors Create Operational and Economic Risk:

Grid Imbalance

Inaccurate forecasts complicate supply–demand balancing.

Poor Scheduling

Operators cannot plan dispatch and storage efficiently.

Higher Costs

Forecast errors may increase reserve requirements and energy waste.

Model requirement

Panel degradation and changing operating conditions affect generation over time.

The model must capture patterns across multiple time scales.

Problem Definition: Existing Research Landscape

Study	Dataset / Horizon	Main Limitation	Model	Reported Result
Konstantinou et al. (2021)	Cyprus PV plant 1.5-hour ahead	Single plant and limited horizon	Stacked LSTM	RMSE = 0.11368
Jesus (2025)	BR-PVGen 51 plants 15 Min ahead	Static models limited temporal learning	MLP + TL / RF	SMAPE = 43.92% NRMSE = 14.07% / 13.82%
Radzi et al. (2026)	Malaysian PV plant 38 months	Higher complexity and deployment cost	Hybrid GRU– LSTM	$R^2 = 0.9961$ RMSE = 18.55 W MAE = 19.43 W

Results are not directly comparable because datasets, forecast horizons, units, and evaluation protocols differ.

These differences motivate a consistent multi-plant, multi-horizon evaluation.

Problem Definition: Research Gap

Generalization Across Plants and Horizons Remains Difficult:

- Many studies focus on a single PV plant or a limited operating condition.
- A model that performs well at 15 minutes may not perform equally well at 24 hours.
- Results are difficult to compare because datasets, forecast horizons, units, and evaluation protocols differ.

A consistent multi-plant, multi-horizon evaluation is still needed.

Motivation & Objectives: Why Machine Learning ?

Machine Learning Learns Complex Weather–Power Relationships:

Nonlinear Modeling

Captures complex relationships between weather, solar position, and PV output.

Multi-Source Features

Combines weather data, historical power, solar geometry, and plant metadata.

Multi-Horizon Forecasting

Supports 15-minute-ahead and 24-hour-ahead solar power prediction.

Operational Value

Supports energy scheduling, grid integration, and performance monitoring.

Machine learning transforms heterogeneous solar data into actionable power forecasts.

Motivation & Objectives: Project Rationale

Two Complementary Models Need a Fair Comparison:

Random Forest provides an interpretable and robust baseline with feature-importance analysis.

LSTM models sequence order and long-term temporal dependencies.

The project compares both models across 51 PV plants and two operationally relevant horizons: 15 minutes and 24 hours.

Evaluation question: Does explicit temporal learning improve forecasting consistently across horizons?

Motivation & Objectives: Project Goals

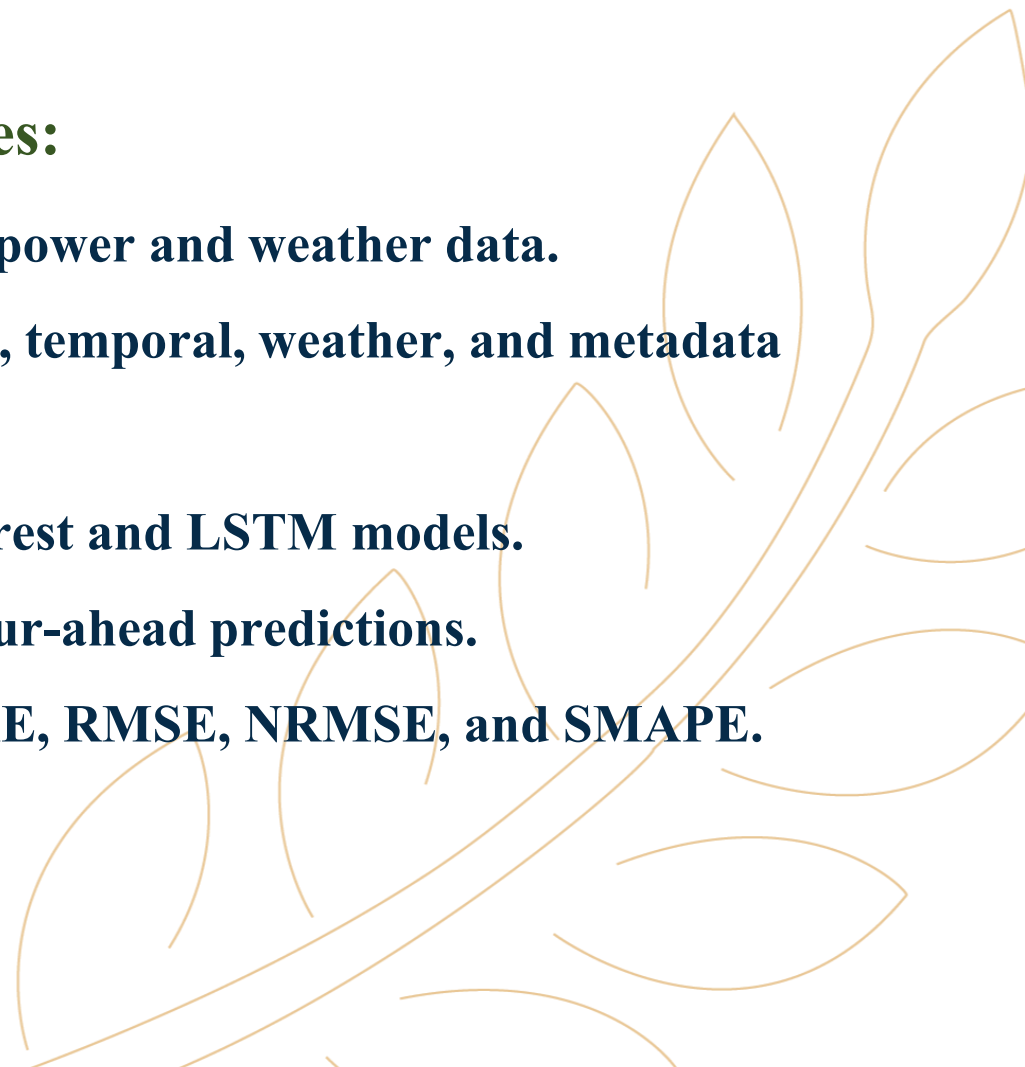
The Project Targets Reliable Multi-Horizon Forecasting:

- **Enable smarter energy trading decisions by providing accurate day-ahead solar power forecasts that allow plant operators and grid managers to reduce reserve costs and maximise revenue from electricity market participation**
- **Design a generalised forecasting framework that adapts to new unseen plants using plant metadata (capacity, efficiency, tracker type, location) without requiring retraining from scratch**
- **Implement Transfer Learning to improve prediction accuracy for plants with limited historical data by leveraging knowledge from data-rich plants**
- **Deploy the trained models through a web application that allows users to input plant specifications and weather forecasts and receive predicted power output curves for any plant — known or new**

Implementation: End-to-End Pipeline



The Proposed Solution Follows Five Connected Stages:

1. **Data Preparation:** Clean, synchronize, pair, and resample power and weather data.
 2. **Feature Engineering:** Generate solar geometry, lag, rolling, temporal, weather, and metadata features.
 3. **Model Training:** Train independent per-plant Random Forest and LSTM models.
 4. **Forecast Generation:** Produce 15-minute-ahead and 24-hour-ahead predictions.
 5. **Evaluation & Comparison:** Compare models using R^2 , MAE, RMSE, NRMSE, and SMAPE.
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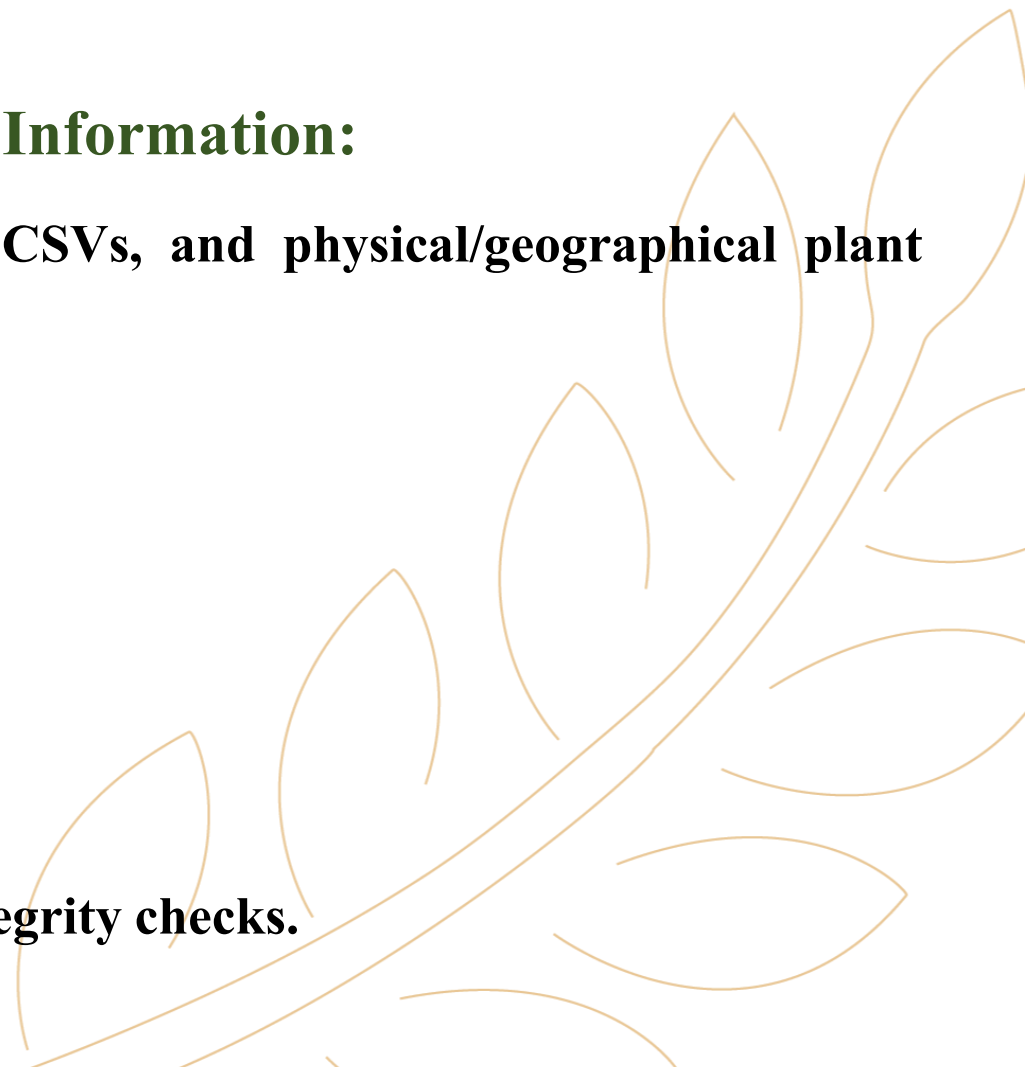
Implementation: Data Preparation



Data Preparation Aligns Power, Weather, and Plant Information:

Raw Inputs include power CSVs from 51 plants, weather CSVs, and physical/geographical plant metadata.

Processing Steps:

- **Pair power and weather records by plant identity.**
 - **Validate plant folders, profiles, and metadata.**
 - **Normalize time zones and align timestamps to UTC.**
 - **Merge power and weather data at 15-minute intervals.**
 - **Interpolate missing values where appropriate and apply integrity checks.**
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Implementation: Feature Engineering



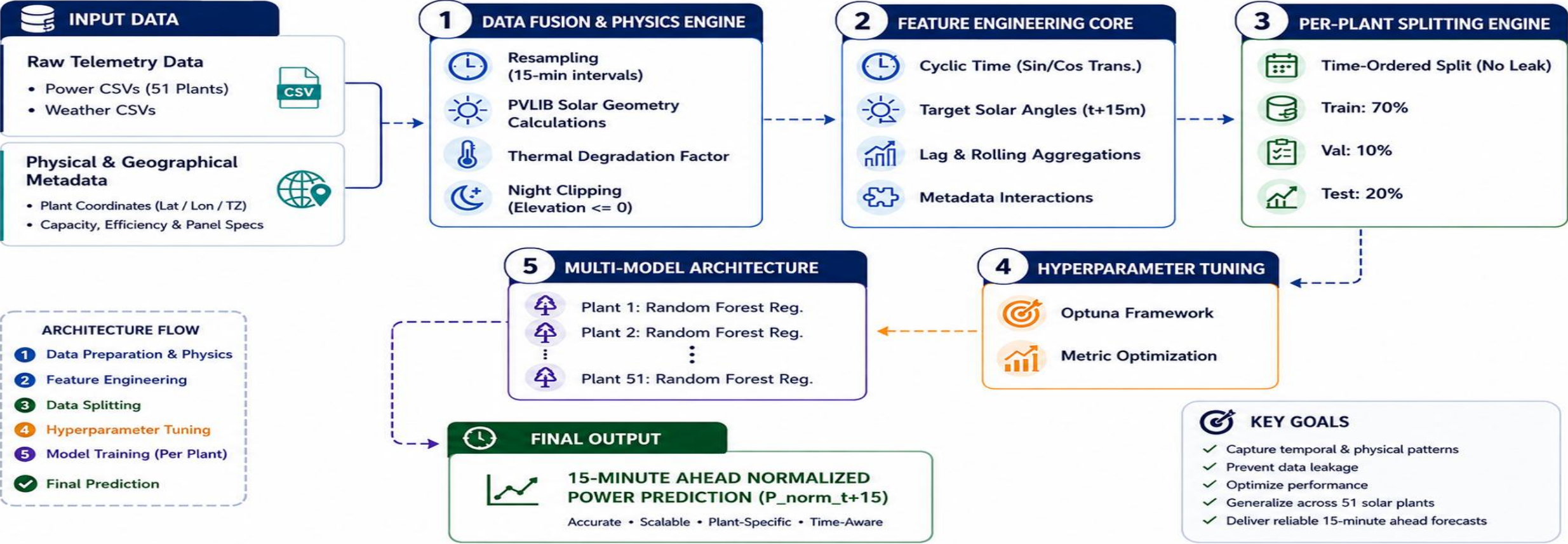
Physics-Informed Features Represent Solar Behavior:

- **Solar Geometry:** elevation, azimuth, zenith, clear-sky index, and geometry interactions.
 - **Temporal Features:** hour, day, season, and cyclic sine/cosine transformations.
 - **Lag Features:** previous PV generation and lagged meteorological variables.
 - **Rolling Features:** short-term statistics and temporal variability indicators.
 - **Plant Metadata:** tracker type, bifaciality, installed capacity, location, and panel characteristics.
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Architecture: 15-Min Ahead RF

SOLAR POWER FORECASTING SYSTEM (15-MINUTE AHEAD)

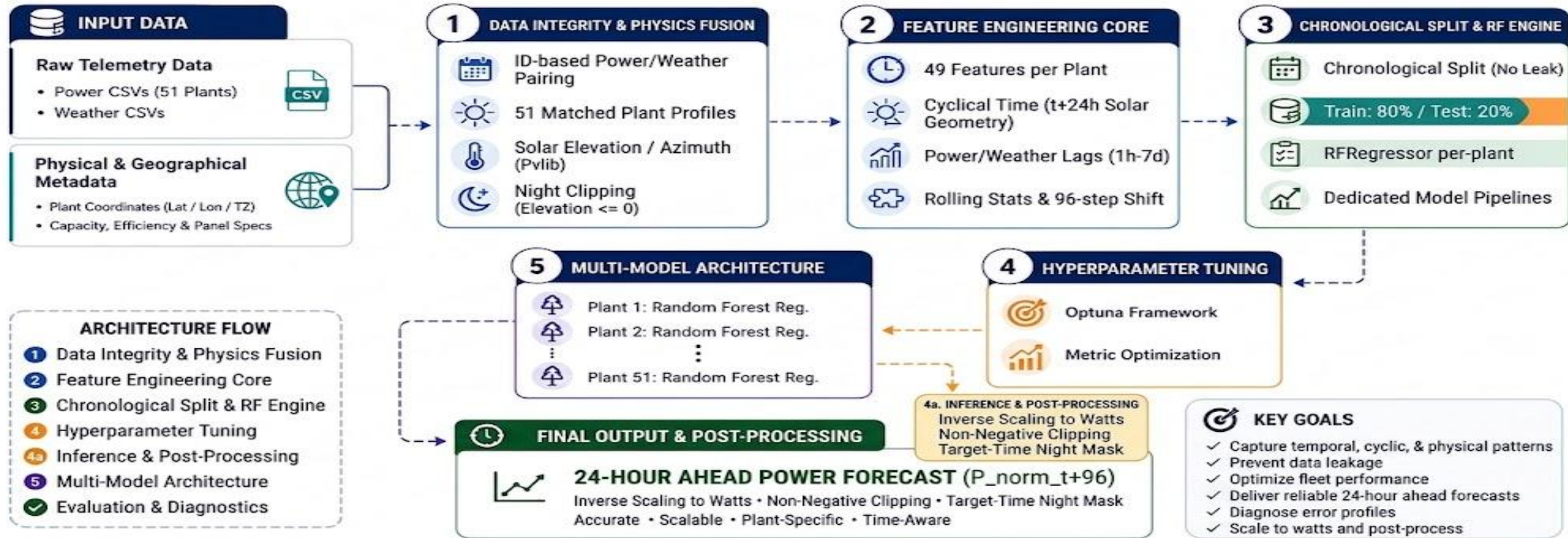
RF-Architecture



Architecture: 24-H Ahead RF

SOLAR POWER FORECASTING SYSTEM (24-HOUR AHEAD)

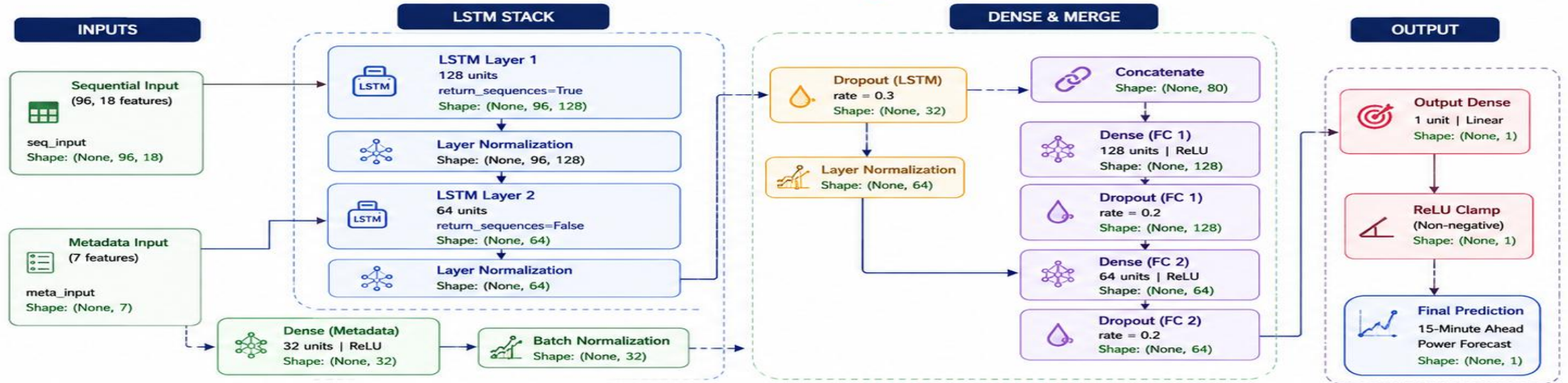
RF-Architecture



Architecture: (LSTM) 15-min/24-H Ahead

LSTM ARCHITECTURE

Time Series Forecasting Model



Layer (type)	Output Shape	Param #	Connected to
seq_input (InputLayer)	(None, 96, 18)	0	-
meta_input (InputLayer)	(None, 7)	0	-
lstm_1 (LSTM)	(None, 96, 128)	75,264	seq_input[0][0]
meta_dense_1 (Dense)	(None, 32)	256	meta_input[0][0]
layer_normalization (LayerNormalization)	(None, 96, 128)	256	lstm_1[0][0]
batch_normalization (BatchNormalization)	(None, 32)	128	meta_dense_1[0][0]
lstm_2 (LSTM)	(None, 64)	49,408	layer_normalization[0][0]
dropout (Dropout)	(None, 32)	0	batch_normalization[0][0]
layer_normalization_1 (LayerNormalization)	(None, 64)	128	lstm_2[0][0]

Layer (type)	Output Shape	Param #	Connected to
meta_dense_2 (Dense)	(None, 16)	528	dropout[0][0]
merge (Concatenate)	(None, 80)	0	layer_normalization_1[0][0], meta_dense_2[0][0]
fc_1 (Dense)	(None, 128)	10,368	merge[0][0]
dropout_1 (Dropout)	(None, 128)	0	fc_1[0][0]
fc_2 (Dense)	(None, 64)	8,256	dropout_1[0][0]
dropout_2 (Dropout)	(None, 64)	0	fc_2[0][0]
output (Dense)	(None, 1)	65	dropout_2[0][0]
relu_clamp (ReLU)	(None, 1)	0	output[0][0]

NOTES:

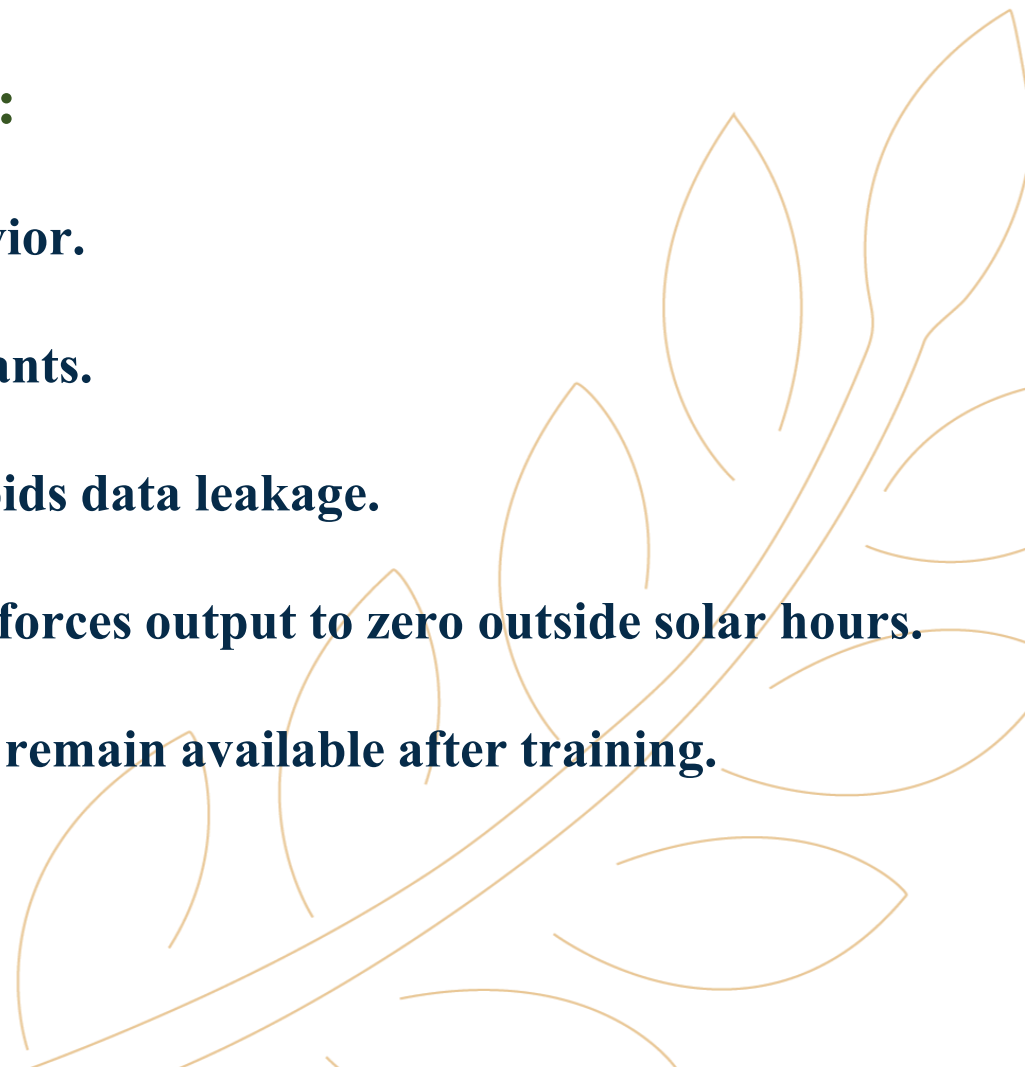
- None = Batch Size (variable)
- All Dense layers use ReLU activation except Output (Linear)
- Dropout rates: LSTM=0.3, FC=0.2

TOTAL TRAINABLE PARAMETERS: 145,001

Methodology: Training Strategy



Per-Plant Modeling Preserves Site-Specific Behavior:

- **One model per plant captures site-specific generation behavior.**
 - **Shared feature schema keeps inputs consistent across all plants.**
 - **Chronological evaluation preserves temporal order and avoids data leakage.**
 - **Night-time hard masking applies a physical constraint that forces output to zero outside solar hours.**
 - **Operational benefit: plant-level forecasting and diagnostics remain available after training.**
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Methodology: Random Forest Configuration

RF Hyperparameters Are Tuned Separately for Each Horizon:

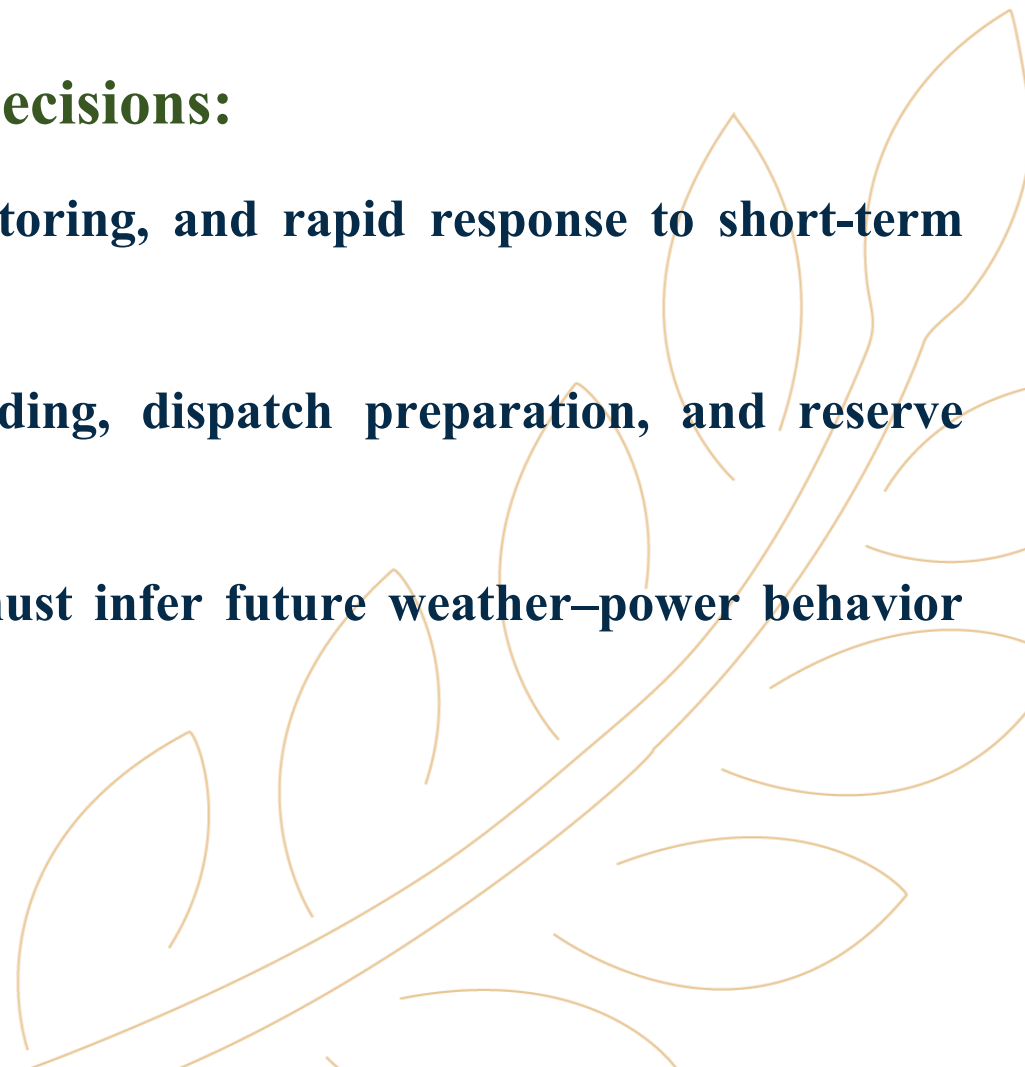
Parameter	15-Minute Experiment	24-Hour Experiment
n_estimators	422	150
max_depth	18	10
min_samples_split	4	30
min_samples_leaf	11	12
max_features	0.3	sqrt

Tuning objective: optimize the selected evaluation metric while controlling model complexity and overfitting.

Methodology: Forecast Horizons



The Two Horizons Represent Different Operational Decisions:

- **15-Minute Ahead** supports near-real-time scheduling, monitoring, and rapid response to short-term changes.
 - **24-Hour Ahead** supports day-ahead planning, energy trading, dispatch preparation, and reserve decisions.
 - The longer horizon is more uncertain because the model must infer future weather–power behavior further from the latest observation.
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Methodology: Evaluation Protocol

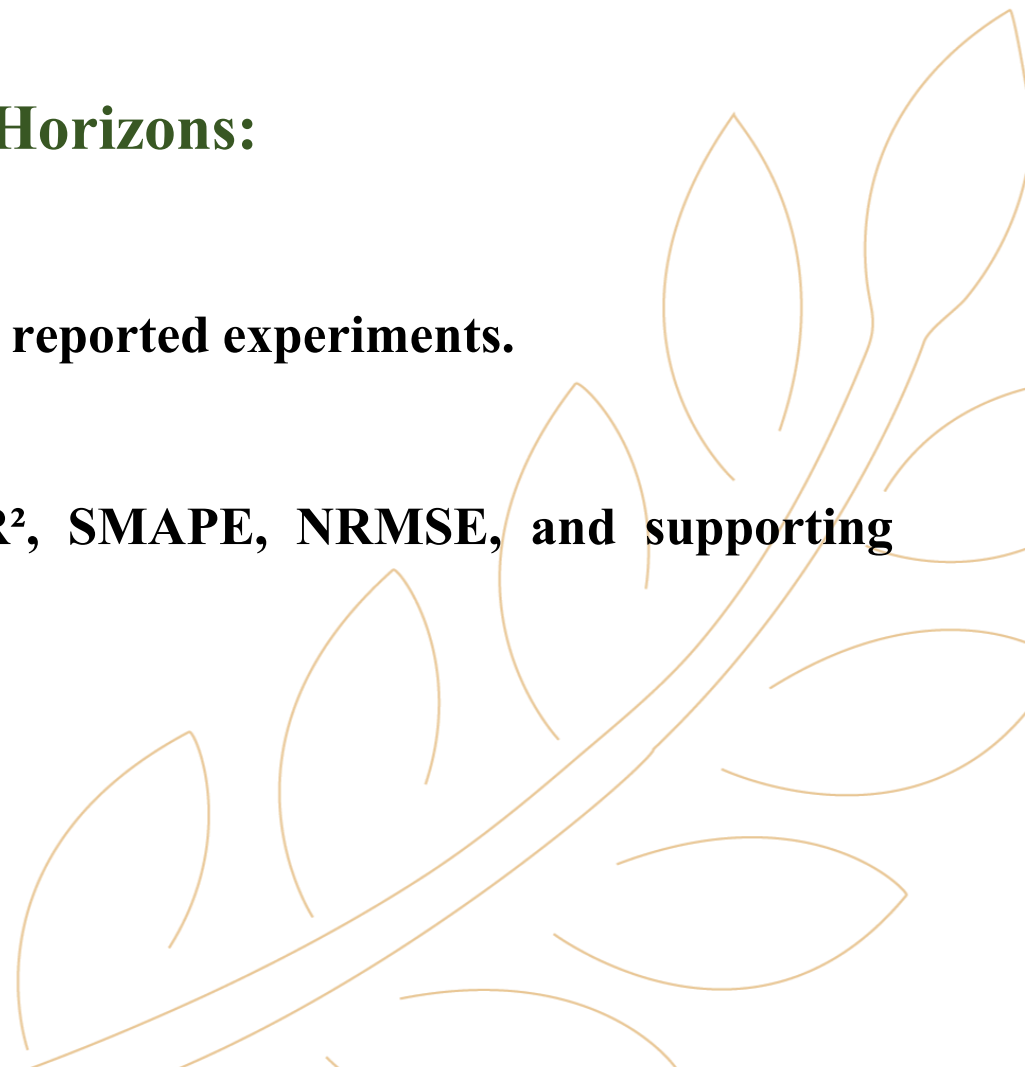
Evaluation Combines Accuracy, Error, and Diagnostics:

- **Accuracy:** R^2 measures goodness of fit.
- **Error:** MAE, RMSE, NRMSE, and SMAPE quantify forecast deviation.
- **Structure:** metrics are reviewed at both pooled fleet level and per-plant level.
- **Diagnostics:** prediction curves, outlier detection, feature importance, night behavior, and error profiles explain when and why performance changes.
- **Reported scope:** 50 plants were evaluated after excluding Plant 15 as a severe outlier.

Results: Evaluation Overview



The Experiments Compare Two Models Across Two Horizons:

- **Dataset:** BR-PVGen with 51 PV plants.
 - **Evaluated plants:** 50 plants after excluding Plant 15 from the reported experiments.
 - **Forecast horizons:** 15 minutes ahead and 24 hours ahead.
 - **Main comparison:** Random Forest versus LSTM using R^2 , SMAPE, NRMSE, and supporting diagnostics.
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Results: Random Forest, 15 Minutes

Random Forest Delivers Strong Short-Horizon Error Metrics:

Metric	Result
Mean R ²	0.8883
Mean SMAPE	17.81%
Mean NRMSE	10.52%
Plants evaluated	50

Insights: Predictions closely follow actual power, most plants achieved R² above 0.80, and weather plus historical power were the most influential features.

Limitation: Performance decreases during rapid weather changes because RF has no explicit temporal memory.

Results: Random Forest, 24 Hours

The 24-Hour Horizon Is More Difficult for Random Forest:

Metric	Result
Mean R ²	0.7820
Mean SMAPE	22.50%
Mean NRMSE	15.29%
Plants evaluated	50

Insights: Several plants exceeded R² above 0.85, while plant-level performance was heterogeneous and Plant 15 was a severe outlier.

Limitation: The longer horizon increases uncertainty and exposes the absence of explicit temporal dynamics.

Results: LSTM, 15 Minutes

LSTM Achieves the Highest Mean R^2 at 15 Minutes:

Metric	Result
Mean R^2	0.9050
Mean SMAPE	36.88%
Mean NRMSE	15.82%
Plants evaluated	50

Insights: LSTM captures the main daily generation trend and explicitly models temporal dependencies.

Limitation: Errors increase during rapid weather changes, and the error metrics are higher than the reported RF results in this experiment.

Results: LSTM, 24 Hours

LSTM Captures the Daily Pattern but Smooths Sharp Peaks:

Metric	Result
Mean R ²	0.7919
Mean SMAPE	33.30%
Mean NRMSE	29.64%
Plants evaluated	50

- Insights:** The model captures the broad solar-generation pattern and achieves a mean R² of approximately 0.79.
- Limitation:** It underestimates rapid changes, shows higher errors around sunrise and sunset, and requires 192 historical timesteps.

Results: Cross-Model Comparison

Model Selection Depends on the Metric and Horizon:

Model	Horizon	Mean R ²	SMAPE (%)	NRMSE (%)
LSTM	15 min	0.9050	36.88	15.82
RF	15 min	0.8883	17.81	10.52
LSTM	24 h	0.7919	33.30	29.64
RF	24 h	0.7820	22.50	15.29

Overall finding: LSTM records the higher mean R² at both horizons, while Random Forest records lower SMAPE and NRMSE and remains easier to interpret.

" The best model depends on the operational horizon and target performance metric."

Results: Reference Comparison

The Reported RF Results Improve on Listed Reference Errors:

Model	SMAPE (%)	NRMSE (%)
ARIMA	47.05	14.23
Gradient Boosting	46.94	13.99
Random Forest	47.33	13.82
XGBoost	46.48	13.92
MLP	46.40	14.49
MLP + Transfer Learning	43.92	14.07
Our RF: 15 min	17.81%	10.52%
Our RF: 24 h	22.50%	15.29%
Our LSTM: 15 min	36.88%	15.82%
Our LSTM: 24 h	33.30%	29.64%

Key message: Under the comparison protocol used in the presentation, the reported RF results achieve lower SMAPE and NRMSE than the listed 15-minute reference results.

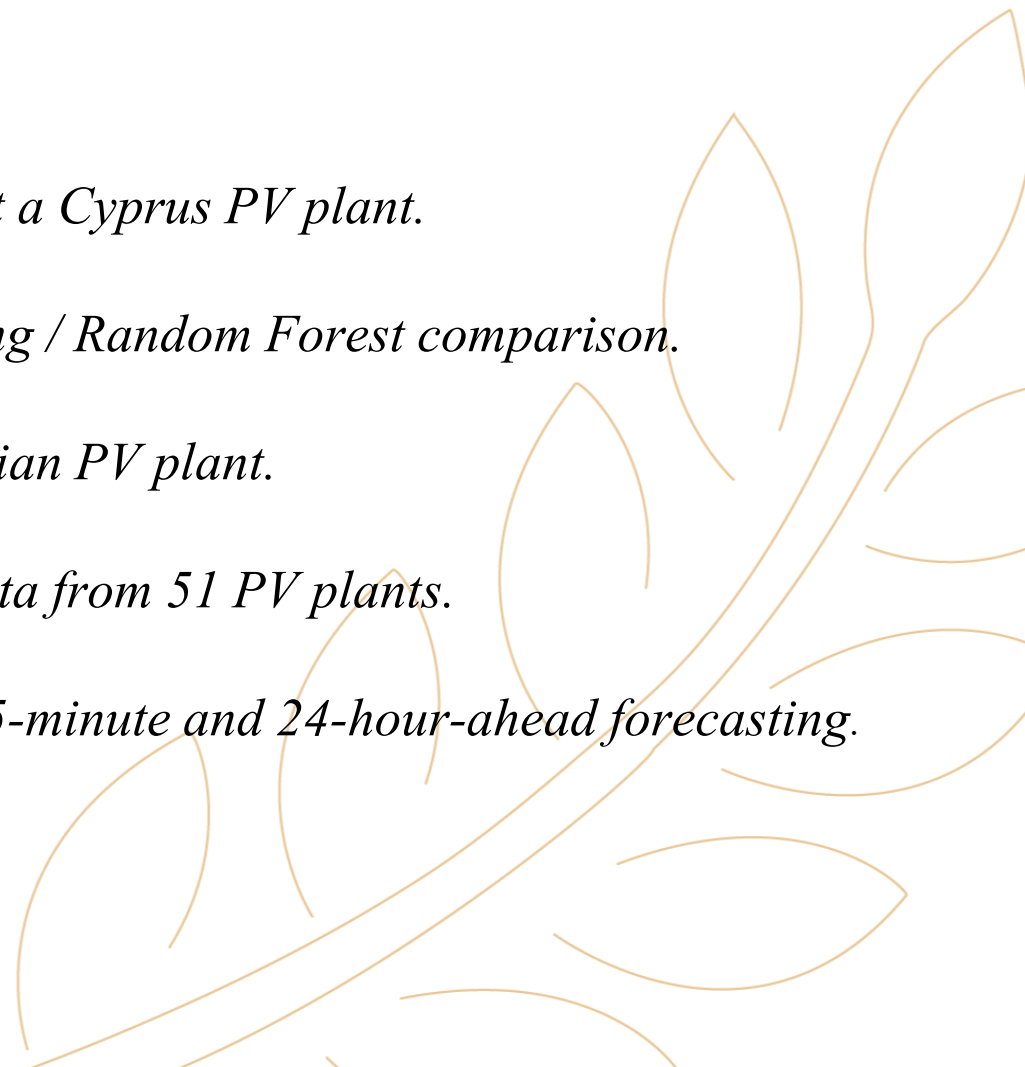
Future Work

The Next Step Is Real-Time Operational Deployment:

- **Hybrid RF + LSTM Models** combine RF interpretability with LSTM temporal learning.
- **Real-Time Forecasting** deploys the pipeline for continuous operational predictions and monitoring.
- **Richer Data and Generalization** add weather variables, evaluate larger datasets, and improve transfer to plants with limited historical data.
- **Scalable Deployment** extends Sunova toward cloud services, smart-grid integration, IoT connectivity, and a user-facing forecasting application.

“ The goal is to turn the forecasting pipeline into a real-time operational tool. ”

References



1. *Konstantinou et al. (2021), Stacked LSTM for solar forecasting at a Cyprus PV plant.*
2. *Jesus (2025), BR-PVGen benchmark and MLP / Transfer Learning / Random Forest comparison.*
3. *Radzi et al. (2026), Hybrid GRU–LSTM forecasting for a Malaysian PV plant.*
4. *BR-PVGen benchmark: meteorological and power-generation data from 51 PV plants.*
5. *Project experiments: Random Forest and LSTM evaluation for 15-minute and 24-hour-ahead forecasting.*



Thank You !

Questions & Discussion

